

- Q.26 With the help of block diagram, explain in brief the data communication.
- Q.27 Write main features of Frequency hopping spectrum.
- Q.28 Differentiate between asynchronous & synchronous transmission.
- Q.29 Draw & explain full duplex circuit.
- Q.30 Write short note on
- a) Noise
 - b) Data transmission speed
- Q.31 Show how PSK is detected?
- Q.32 State & explain sampling theorem, write it's importance.
- Q.33 What is need of Modem in communication system?
- Q.34 Explain the concept of noise in data communication circuit.
- Q.35 Differentiate between circuit & packet switching.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain various modem Modulation methods.
- Q.37 Explain PAM, PPM, and PWM.
- Q.38 Explain in detail the Delta Modulation. Draw all diagrams.

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5th Sem / Eltx
Subject:- Digital Communication

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Sampling theorem is used for converting_____
- a) Continuous time signal to discrete
 - b) Discrete to continuous signal
 - c) Continuous time signal to discrete & vice versa
 - d) None of these
- Q.2 The factors that cause quantizing error in delta modulation are:
- a) Slope over load distortion
 - b) Granular Noise
 - c) A & B both
 - d) None
- Q.3 Which pulse modulation technique is least expensive?
- a) PAM
 - b) PPM
 - c) PWM
 - d) PCM

- Q.4 Which one of the following is analog?
 a) PCM b) PWM
 c) Delta Modulation d) Differential PCM
- Q.5 MODEM stands for
 a) Modifying Electromagnetic Waves
 b) Delta Modulation
 c) Modulator=Demodulator
 d) None
- Q.6 What is DSL in Modem?
 a) Dual Signal line
 b) Digital subscriber line
 c) Digital signal layer
 d) Dual signal layer
- Q.7 The time switch has a memory cycle at _____ MHz.
 a) 833 b) 842
 c) 853 d) 862
- Q.8 APCM frame contains a group of _____ bits.
 a) 16 b) 24
 c) 28 d) 32
- Q.9 Crosstalk is expressed in
 a) dBm b) dBfs
 c) dB d) Watts

- Q.10 The advantage of space switching is that is _____?
 a) Is instantaneous b) is error free
 c) Need no flow control d) is free from blocking

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write any two advantages of digital communication.
 Q.12 Define DPCM?
 Q.13 Define crosstalk?
 Q.14 Define aperture effect.
 Q.15 Define data communication?
 Q.16 Expand UART and USART
 Q.17 Draw a simplex circuit
 Q.18 Define equalizers?
 Q.19 Define Modem interfacing
 Q.20 STS _____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a short note on working principle of ASK
 Q.22 Explain the concept of COMPANDING
 Q.23 Explain working principle of TST switch.
 Q.24 Explain various modes of modem Operations
 Q.25 Compare analog and digital communication systems